PHLO
JACKET

Apparel Studio
SPD 685

ELLIOT HILDNER



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Oregon State University

B.S. Mechanical Engineering

UNIVERSITY OF OREGON

M.S. Sports Product Design (First-Year)



ENVIRONMENT

Rain Is Part of the Game

- Golfers play through **rain, wind**, and **cold**
- Rounds continue for **hours** in wet conditions
- **Water travels** down the body, not just onto it
- Moisture directly affects **grip, mobility**, and **comfort**

THE ATHLETE

“FOR THE LOVE OF THE GAME“

THE PROBLEM

WET GLOVES

→ Cold, water-logged hands lose grip and feel, directly hurting performance

SOAKED THIGHS

→ Rain travels off the torso and dumps onto the legs, adding weight and discomfort

RESTRICTED MOBILITY

→ Rain jackets stiffen or cling when wet, limiting rotation and follow-through



MARKET LANDSCAPE

STRENGTHS

- Waterproof fabrics
- Stretchy + breathable
- Wrist + waist cinches
- Clean, athletic silhouettes





GAPS

- No control of water flow
- No solutions for hand dryness
- Mobility still compromised

JOBS TO BE DONE



MAINTAIN RELIABLE GRIP AND HAND FUNCTION IN WET CONDITIONS



MAINTAIN MOBILITY AND SWING FREEDOM IN WET CONDITIONS



PREVENT MOISTURE ACCUMULATION ON FOREARMS AND SLEEVES



PREVENT RAINWATER FROM RUNNING ONTO THE LEGS



TRADE-OFFS

A golf rain jacket must balance
water protection, mobility, and **comfort**

DESIGN CONSIDERATIONS

- **Protection** without losing feel
- **Mobility** without restriction
- **Dry hands** without bulky solutions
- **Comfort** without sacrificing performance

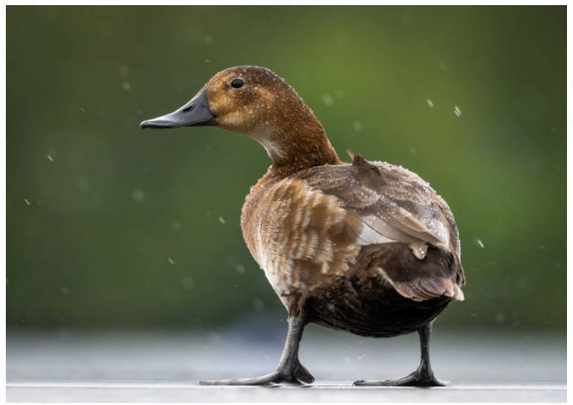


How might we design a golf rain jacket using biomimicry that keeps the golfer's hands and thighs dry, while maintaining full swing flexibility in wet conditions?

BIOMIMICRY INSIGHT

Nature uses **surface geometry,** micro-textures, and directional structures to shed and **guide water** efficiently









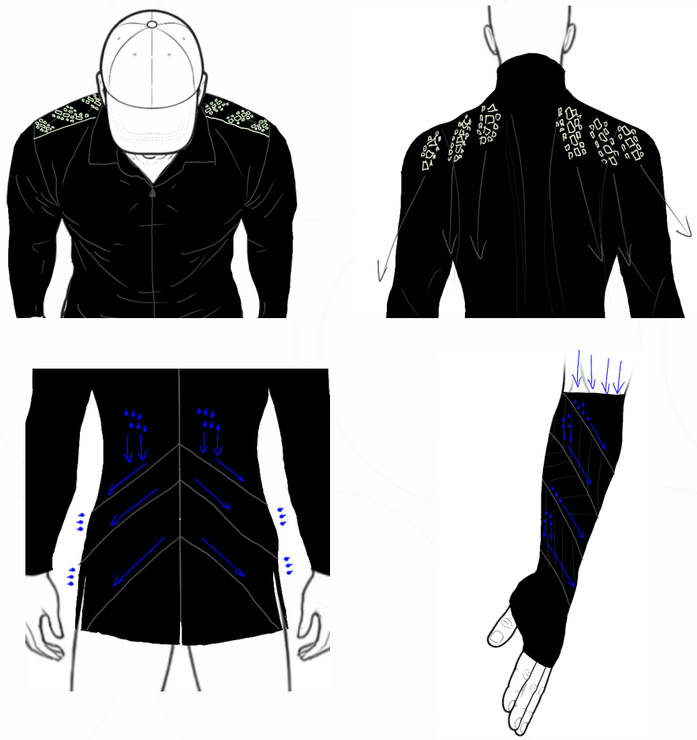


DESIGN EVOLUTION

Researching **natural** and **technical strategies** and translating them into **three** concept directions

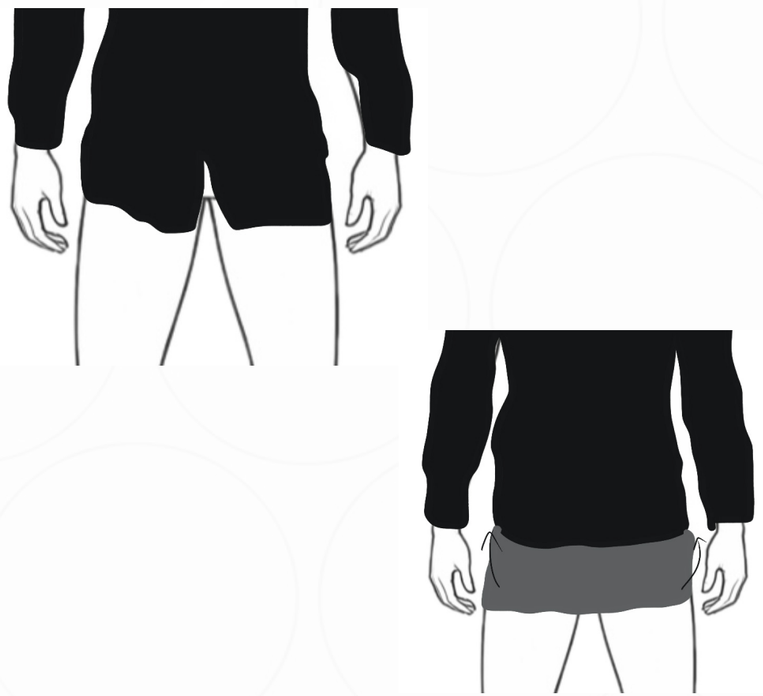
Biomimicry Water-Flow Control

- Crocodile directional texture
- **Leaf gutter system**



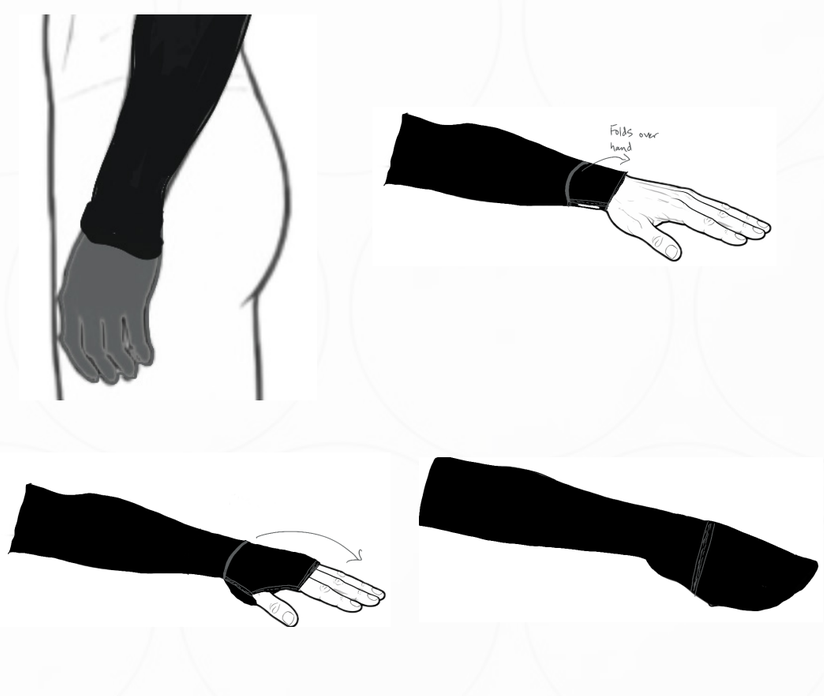
Waist and Thigh Dryness

- **Extended split front hem**
- Drop-down thigh shield



Glove–Sleeve Interface Systems

- Internal wrist gaiter
- **Over-glove cover**



PROTOTYPES

1. FIT
2. DETAIL
3. FULL

Validate **movement** and **coverage**









Integrating **channel system** + water-management features



Test **over-glove** and **channel placement**



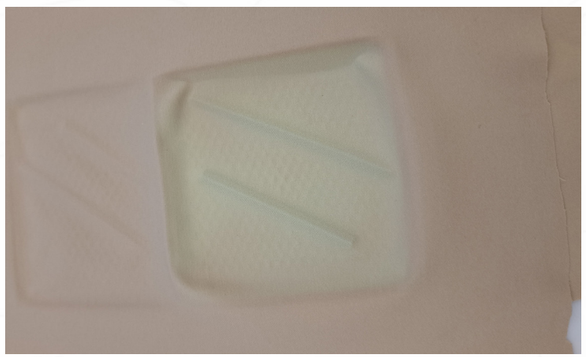
TESTING & VALIDATION

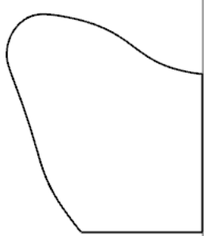
Evaluating **channel constructions** and **water-flow** performance through **material** and **on-body** testing

3D PRINTED TPU

Vacuum-formed strips



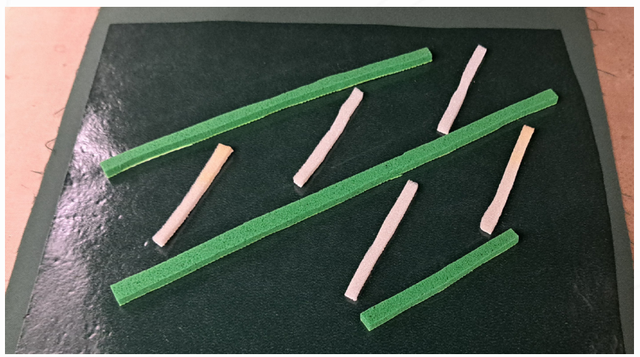




OPEN-CELL FOAM

Heat-pressed strips





WATER TESTING

Faucet + wear tests





MATERIALS & MANUFACTURING

FABRIC

- Recycled Polyester / Lycra
- DWR Stretch Woven
- Bluesign Approved





CHANNEL SYSTEM

- Ortholite Foam (main ribbing)
- Open-Cell Foam w/ Mesh Backing (supporting ribbing)
- Bemis 3914 Adhesive Glue
- Heat-Pressed at 300°F for 30 seconds



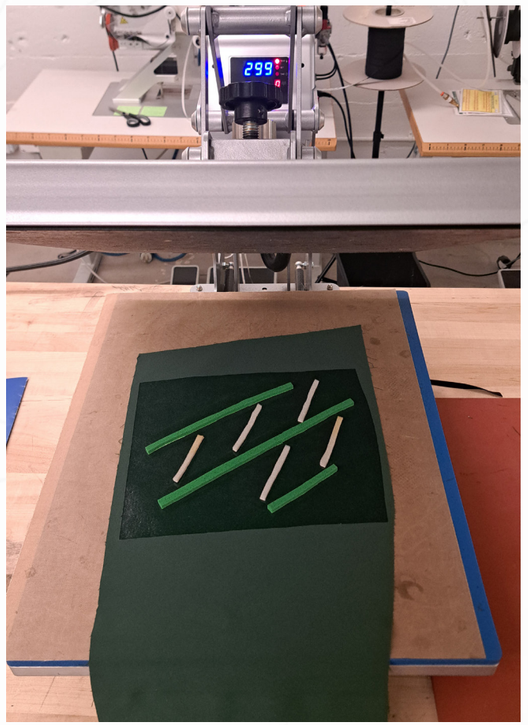
Ortholite Foam



Open Cell Foam
Mesh Backing



Adhesive Glue



PHLO JACKET

Collar to shield neck from rain

Integrated **over-mitt sleeves** to shield the glove/hands between shots





Integrated **foam-raised channels** to direct waterflow off key zones (forearms, back, waist)







Dual **side zippers** for increased mobility during swing

Dual-direction front zipper to allow venting and fit adjustment